

Total No. of Printed Pages: 1

SUBJECT CODE NO: RNP-8006
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (CSE/CS/CE/AI&DS/AIML/IT/AI) (NEP) Sem - III
Examination May/June-2026
Data Structures

[Time: 3:00 Hours]

[Max. Marks: 60]

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from question no's 2, 3, 4, 5, 6 and 7

SECTION-A

Q.1 Solve any TEN from the following questions:

20

- a) Define array. Give two examples.
- b) What is pointer?
- c) Define Big-O notation.
- d) Write two applications of a queue.
- e) What is a dequeue?
- f) Define doubly linked list.
- g) What is a binary tree?
- h) What is Binary Search Tree (BST)?
- i) State two differences between insertion sort and selection sort.
- j) What is hashing?
- k) Define graph.
- l) What is BFS traversal?
- m) Define heap sort.

SECTION-B

Q.2 a) Explain different data structure operations with suitable examples. **05**

b) Explain stack using array and its operations. **05**

Q.3 a) Explain circular queue with example. **05**

b) Write algorithm for insertion and deletion in linked list. **05**

Q.4 a) Explain binary tree representation in memory. **05**

b) Explain tree traversal techniques. **05**

Q.5 a) Explain merge sort with example. **05**

b) Compare different sorting techniques. **05**

Q.6 a) Explain linear and binary search. **05**

b) Explain hashing and hash functions. **05**

Q.7 a) Explain DFS traversal with example. **05**

b) Explain Kruskal's algorithm. **05**